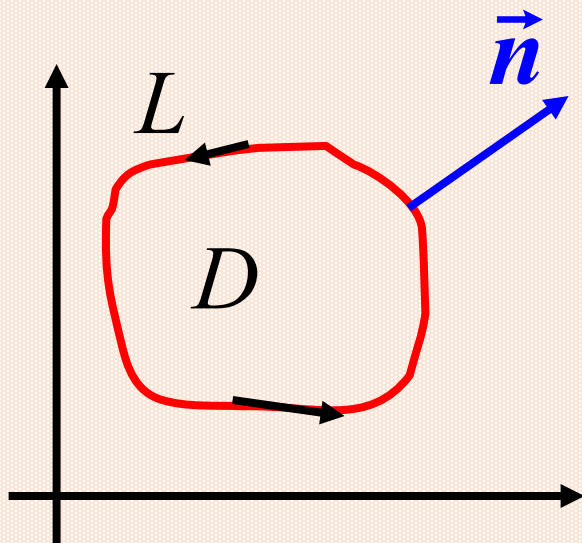


## 专题：Green公式、Gauss公式的特殊情形

$$\oint_L \mathbf{P}dx + \mathbf{Q}dy = \iint_D \left( \frac{\partial \mathbf{Q}}{\partial x} - \frac{\partial \mathbf{P}}{\partial y} \right) dxdy$$

格林公式的另一形式：用第一型曲线积分表示

$$\oint_L (P \cos \alpha + Q \cos \beta) ds = \iint_D \left( \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dxdy$$

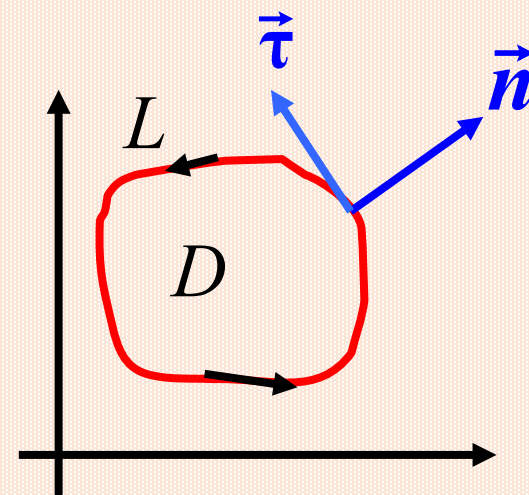


$$\vec{n} = \{\cos \alpha, \cos \beta\}$$

$$\iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_L P dx + Q dy$$

**推导** 将  $Q$  换成  $P$ ,  $P$  换成  $-Q$ , 得

$$\begin{aligned} \iint_D \left( \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy &= \oint_L -Q dx + P dy \\ &= \oint_L \{-Q, P\} \cdot \tau ds \\ &= \oint_L \{-Q, P\} \cdot \{-\cos \beta, \cos \alpha\} ds \\ &= \oint_L \{P, Q\} \cdot \{\cos \alpha, \cos \beta\} ds \end{aligned}$$



$$\vec{n} = \{\cos \alpha, \cos \beta\} \quad \vec{\tau} = \{\cos \theta_x, \cos \theta_y\} = \{-\cos \beta, \cos \alpha\}$$

特别的我们有

$$\oint_L \frac{\partial u}{\partial \vec{n}} ds = \oint_L \left( \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta \right) ds$$
$$= \iint_D \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy$$

$$\begin{aligned}
\oint_L v \frac{\partial u}{\partial \vec{n}} ds &= \oint_L \left( v \frac{\partial u}{\partial x} \cos \alpha + v \frac{\partial u}{\partial y} \cos \beta \right) ds \\
&= \iint_D \left( \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial y^2} \right) dx dy \\
&= \iint_D (\nabla u \cdot \nabla v) dx dy + \iint_D v \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy
\end{aligned}$$

特别地

$$\oint_L u \frac{\partial u}{\partial \vec{n}} ds = \iint_D |\nabla u|^2 dx dy + \iint_D u \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy$$

**例1** 计算  $\oint_L \frac{\partial f}{\partial \vec{n}} ds$ , 其中  $L$  为椭圆  $2x^2 + y^2 = 1$ ,  
 $f(x, y) = (x-2)^2 + y^2$ ,  $\vec{n}$  为  $L$  的单位外法向量

**解** 设  $L$  围成的区域为  $D$ , 由 Green 公式

$$\begin{aligned}\oint_L \frac{\partial f}{\partial \vec{n}} ds &= \oint_L \left( \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta \right) ds \\ &= \iint_D \left( \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dx dy \\ &= 4 \iint_D dx dy = 4\pi \cdot \frac{1}{\sqrt{2}} \cdot 1 = 2\sqrt{2}\pi.\end{aligned}$$

**例2** 设 $D$ 为平面有界闭区域,  $L$ 为其光滑边界, 函数 $u(x, y)$ 在 $\bar{D}$ 上连续, 在 $D$ 内有连续二阶偏导, 且在 $D$ 内  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ , 在 $L$ 上  $u(x, y) = 0$ . 证明:  $u(x, y) \equiv 0, (x, y) \in D$ .

**证** 由Green公式,

$$\oint_L u \frac{\partial u}{\partial \vec{n}} ds = \iint_D |\nabla u|^2 dx dy + \iint_D u \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy$$

其中,  $\vec{n}$  表示 $L$ 单位外法向量. 于是, 由已知条件有

$$\iint_D |\nabla u|^2 dx dy = 0. \text{ 于是, } \nabla u \equiv 0, (x, y) \in D. \text{ 故 } u \equiv C, (x, y) \in D.$$

由于在 $L$ 上  $u(x, y) = 0$ , 故  $u \equiv 0, (x, y) \in D$ .

**例3** 设  $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$ ,  $f(x, y)$  在  $D$  上二次连续可微,

且满足  $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^{-x^2-y^2}$ , 证明:  $\iint_D \left( x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) dx dy = \frac{\pi}{2e}$ .

**证** 
$$\iint_D \left( x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) dx dy = \int_0^1 r dr \int_0^{2\pi} r \left( \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta \right) d\theta,$$

注意到, 对于固定的  $r \in [0, 1]$ , 曲线  $C_r : x^2 + y^2 = r^2$  的参数方程为

$C_r : x = r \cos \theta, y = r \sin \theta, 0 \leq \theta \leq 2\pi$ , 故,

$$\int_0^{2\pi} r \left( \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta \right) d\theta = \oint_{C_r} \left( \frac{\partial f}{\partial x} \cdot \frac{x}{r} + \frac{\partial f}{\partial y} \cdot \frac{y}{r} \right) ds,$$



$$\int_0^{2\pi} r \left( \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta \right) d\theta = \oint_{C_r} \left( \frac{\partial f}{\partial x} \cdot \frac{x}{r} + \frac{\partial f}{\partial y} \cdot \frac{y}{r} \right) ds$$

注意到,  $C_r$ 上任一点 $(x, y)$ 的外法向量的方向余弦

$$(\cos \alpha, \cos \beta) = \left( \frac{x}{r}, \frac{y}{r} \right), \text{ 故,}$$

$$\int_0^{2\pi} r \left( \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta \right) d\theta = \oint_{C_r} \left( \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta \right) ds$$

$$= \oint_{C_r} \frac{\partial f}{\partial \vec{n}} ds = \iint_{x^2+y^2 \leq r^2} \left( \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dxdy$$

$$= \iint_{x^2+y^2 \leq r^2} e^{-x^2-y^2} dxdy = \int_0^{2\pi} d\varphi \int_0^r e^{-u^2} u du = \pi(1 - e^{-r^2}),$$

$$\iint_D \left( x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) dxdy = \pi \int_0^1 r(1 - e^{-r^2}) dr = \frac{\pi}{2e}.$$

另证 由Green公式

$$\int_{\partial D} -\frac{\partial f}{\partial y} dx + \frac{\partial f}{\partial x} dy = \iint_D \left( \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dx dy = \iint_D e^{-x^2-y^2} dx dy = \pi \left( 1 - \frac{1}{e} \right).$$

于是,

$$\int_{\partial D} -(x^2 + y^2) \frac{\partial f}{\partial y} dx + (x^2 + y^2) \frac{\partial f}{\partial x} dy = \int_{\partial D} -\frac{\partial f}{\partial y} dx + \frac{\partial f}{\partial x} dy = \pi \left( 1 - \frac{1}{e} \right).$$

另一方面,

$$\int_{\partial D} -(x^2 + y^2) \frac{\partial f}{\partial y} dx + (x^2 + y^2) \frac{\partial f}{\partial x} dy$$

Green公式  $\iint_D (x^2 + y^2) e^{-x^2-y^2} dx dy + \iint_D \left( 2x \frac{\partial f}{\partial x} + 2y \frac{\partial f}{\partial y} \right) dx dy.$

而

$$\iint_D (x^2 + y^2) e^{-x^2 - y^2} dx dy = \int_0^{2\pi} d\theta \int_0^1 r^2 e^{-r^2} \cdot r dr = \pi \left(1 - \frac{2}{e}\right).$$

$$\underbrace{\int_{\partial D} -(x^2 + y^2) \frac{\partial f}{\partial y} dx + (x^2 + y^2) \frac{\partial f}{\partial x} dy}_{\pi(1 - \frac{1}{e})} = \underbrace{\iint_D (x^2 + y^2) e^{-x^2 - y^2} dx dy}_{\pi(1 - \frac{2}{e})} + 2 \iint_D \left( x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) dx dy$$

于是,

$$\iint_D \left( x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) dx dy = \frac{\pi}{2e}.$$

**Gauss公式**

$$\iiint_{\Omega} \left( \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV \\ = \oiint_{\Sigma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS.$$

特别地有

$$\oiint_{\Sigma} \frac{\partial u}{\partial \vec{n}} dS = \oiint_{\Sigma} \left( \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma \right) dS \\ = \iiint_{\Omega} \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz = \iiint_{\Omega} \Delta u dx dy dz$$

$$\text{其中 } \Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}.$$

且有

$$\oiint_{\Sigma} v \frac{\partial u}{\partial \vec{n}} dS = \iiint_{\Omega} (\nabla u \cdot \nabla v) dx dy dz + \iiint_{\Omega} v \Delta u dx dy dz$$

特别地

$$\oiint_{\Sigma} u \frac{\partial u}{\partial \vec{n}} dS = \iiint_{\Omega} |\nabla u|^2 dx dy dz + \iiint_{\Omega} u \Delta u dx dy dz$$

**例4** 设函数  $u(x, y, z)$  在由球面  $\Sigma: x^2 + y^2 + z^2 = 2z$  所包围的闭区域  $\Omega$  上具有二阶连续偏导数, 且满足关系式

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = x^2 + y^2 + z^2,$$

$\vec{n}$  为  $\Sigma$  的外法线方向的单位向量, 求  $\iint_{\Sigma} \frac{\partial u}{\partial \vec{n}} dS$ .

**解** 由 Gauss 公式

$$\begin{aligned} \iint_{\Sigma} \frac{\partial u}{\partial \vec{n}} dS &= \iint_{\Sigma} \left( \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma \right) dS \\ &= \iiint_{\Omega} \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz = \iiint_{\Omega} (x^2 + y^2 + z^2) dx dy dz \\ &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2\cos\varphi} r^2 \cdot r^2 \sin \varphi dr = \frac{32}{15} \pi. \end{aligned}$$

**例5** 设 $\Omega$ 为平面有界闭区域,  $\Sigma$ 为其光滑边界函数 $u(x, y, z)$ 在 $\Omega$ 上连续, 在 $\Omega$ 内有连续二阶偏导, 且在 $\Omega$ 内  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ , 在 $\Sigma$ 上  $u(x, y, z) = 0$ . 证明:  $u(x, y, z) \equiv 0, (x, y, z) \in \Omega$ .

**证** 由Gauss公式,

$$\oiint_{\Sigma} u \frac{\partial u}{\partial \vec{n}} dS = \iiint_{\Omega} |\nabla u|^2 dx dy dz + \iiint_{\Omega} u \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz$$

其中,  $\vec{n}$  表示 $\Sigma$ 单位外法向量. 于是, 由已知条件有

$$\iiint_{\Omega} |\nabla u|^2 dx dy dz = 0. \text{ 于是, } \nabla u \equiv 0, (x, y, z) \in D. \text{ 故 } u \equiv C, (x, y, z) \in D.$$

由于在 $\Sigma$ 上  $u(x, y, z) = 0$ , 故  $u \equiv 0, (x, y, z) \in \Omega$ .

**例6** 设  $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$ ,  $u \in C^2(\Omega)$ , 在  $\partial\Omega$  上  $u \equiv 0$ ,

记  $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ , 求证: 对任意  $\varepsilon > 0$ , 有

$$\iiint_{\Omega} |\nabla u|^2 dx dy dz \leq \varepsilon \iiint_{\Omega} (\Delta u)^2 dx dy dz + \frac{1}{4\varepsilon} \iiint_{\Omega} u^2 dx dy dz.$$

**解** 由 Gauss 公式,

$$\oiint_{\partial\Omega} u \frac{\partial u}{\partial \vec{n}} dS = \iiint_{\Omega} |\nabla u|^2 dx dy dz + \iiint_{\Omega} u \Delta u dx dy dz$$

由题设,  $\oiint_{\partial\Omega} u \frac{\partial u}{\partial \vec{n}} dS = 0$ , 从而有,

$$\iiint_{\Omega} |\nabla u|^2 dx dy dz = - \iiint_{\Omega} u \Delta u dx dy dz.$$



注意到,

$$\varepsilon(\Delta u)^2 + \frac{1}{4\varepsilon} u^2 \geq 2\sqrt{\varepsilon} \frac{1}{2\sqrt{\varepsilon}} |u\Delta u| = |u\Delta u|.$$

所以,

$$\begin{aligned} \iiint_{\Omega} |\nabla u|^2 dx dy dz &= - \iiint_{\Omega} u \Delta u dx dy dz \\ &\leq \iiint_{\Omega} |u \Delta u| dx dy dz \\ &\leq \iiint_{\Omega} \left( \varepsilon(\Delta u)^2 + \frac{1}{4\varepsilon} u^2 \right) dx dy dz \\ &\leq \varepsilon \iiint_{\Omega} (\Delta u)^2 dx dy dz + \frac{1}{4\varepsilon} \iiint_{\Omega} u^2 dx dy dz. \end{aligned}$$